

AKTU Lucknow

MID-TERM EXAMINATION ? 2025

Strength of Materials ? Mid-Term Examination, 2025

Subject Code: ME302 Semester: 3 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Stress and Strain)

1. Derive the relationship between stress, strain and elastic constants. Explain the stress-strain curve for mild steel. [12 Marks]
2. How is Stress and Strain implemented? Give an algorithm / procedure and explain. [12 Marks]
3. How is Stress and Strain implemented? Give an algorithm / procedure and explain. [8 Marks]

SECTION B (Unit 2: Shear Force and Bending Moment Diagrams)

4. Draw the shear force and bending moment diagrams for a simply supported beam carrying a uniformly distributed load and a point load. [12 Marks]
5. State and explain the assumptions in the theory of simple bending. Derive the flexure formula. [12 Marks]
6. Solve a problem related to Shear Force and Bending Moment Diagrams. Show all the steps clearly. [12 Marks]

SECTION C (Unit 3: Bending and Shear Stresses)

7. Derive the expression for shear stress distribution in a rectangular beam section and a circular section. [12 Marks]
8. Derive the expression related to Bending and Shear Stresses and explain each term. [12 Marks]
9. Discuss the engineering applications of Bending and Shear Stresses. [12 Marks]

SECTION D (Unit 4: Torsion of Shafts)

10. Derive the torsion formula for a circular shaft. Explain the polar modulus and stiffness of a shaft. [12 Marks]
11. Derive the expression related to Torsion of Shafts and explain each term. [12 Marks]
12. How is Torsion of Shafts implemented? Give an algorithm / procedure and explain. [8 Marks]

SECTION E (Unit 5: Columns and Struts)

13. Compare Euler's and Rankine's formulas for column design with examples. Explain

slenderness ratio. [12 Marks]

14. Discuss the engineering applications of Columns and Struts. [8 Marks]

15. Discuss the engineering applications of Columns and Struts. [12 Marks]

*** End of Question Paper ***